

DISASTER MANAGEMENT

IMPORTANT CONCEPTS AND TERMINOLOGIES

MODULE - 2



VULNERABILITY

Quantitatively correlated with risk, identifies the damageability

CAPACITY

Reciprocal to vulnerability, tries to minimise the risk

MITIGATION

Some actions taken to help reduce or eliminate long-term risks



13 OCTOBER

INTERNATIONAL DAY
FOR DISASTER RISK REDUCTION

**DISASTER
RISK REDUCTION**

Systematic development and application of policies, strategies and practices to minimize vulnerabilities



Vulnerability is the extent to which a community, structure, services or geographic area is likely to be damaged or disrupted by the impact of a particular hazard.

PHYSICAL VULNERABILITY

- It may be determined by aspects such as population density levels, remoteness of a settlement, the site, design and materials used for critical infrastructure and for housing.
- **Wooden homes are less likely to collapse in an earthquake, but are more vulnerable to fire.**



SOCIAL VULNERABILITY

- It refers to the inability of people, organizations and societies to withstand adverse impacts of hazards.
- **When flooding occurs, some citizens, such as children, elderly and differently-abled, may be unable to protect themselves or evacuate, if necessary.**



Vulnerability is the extent to which a community, structure, services or geographic area is likely to be damaged or disrupted by the impact of a particular hazard.

ECONOMIC VULNERABILITY

- The poor are usually more vulnerable to disasters because they lack the resources to build sturdy structures and put other engineering measures in place to protect themselves from being negatively impacted by disasters.
- Poorer families may live in squatter settlements because they cannot afford to live in safer (more expensive) areas.



ENVIRONMENTAL VULNERABILITY

- Natural resource depletion and resource degradation are key aspects of environmental vulnerability.
- Wetlands near the coasts are sensitive to increasing salinity from sea water, and pollution from storm water runoff containing agricultural chemicals, eroded soils, and destruction of mangroves.



POLITICAL FACTORS

- Lack of coordination among various responsible institutions.
- Corruption and vested interests in administration.
- Absence or lax legislations and policies such as land use planning or building regulations.



ECONOMIC FACTORS

- Insufficient funds for disaster management.
- Poverty/ habitation in endangered areas.
- Unsustainable consumption of natural resources.
- Less diversified Economy.



SOCIO-CULTURAL FACTORS

- Insufficient knowledge amongst masses and decreased awareness.
- Superstition – belief that all natural disasters are acts of god and inevitable.
- Unsustainable farm practices which may result in erosion etc.



VULNERABILITY PROFILE OF INDIA

68% of its cultivable area is vulnerable to droughts.

Many hilly areas of the Himalayas and the Western Ghats regions are at risk from landslides and avalanches.

India is also vulnerable to Chemical, Biological, Radiological and Nuclear (CBRN) emergencies and other man-made disasters.

India is vulnerable, in varying degrees, to a large number of disasters

More than 58.6 per cent of the landmass is prone to earthquakes of moderate to very high intensity.

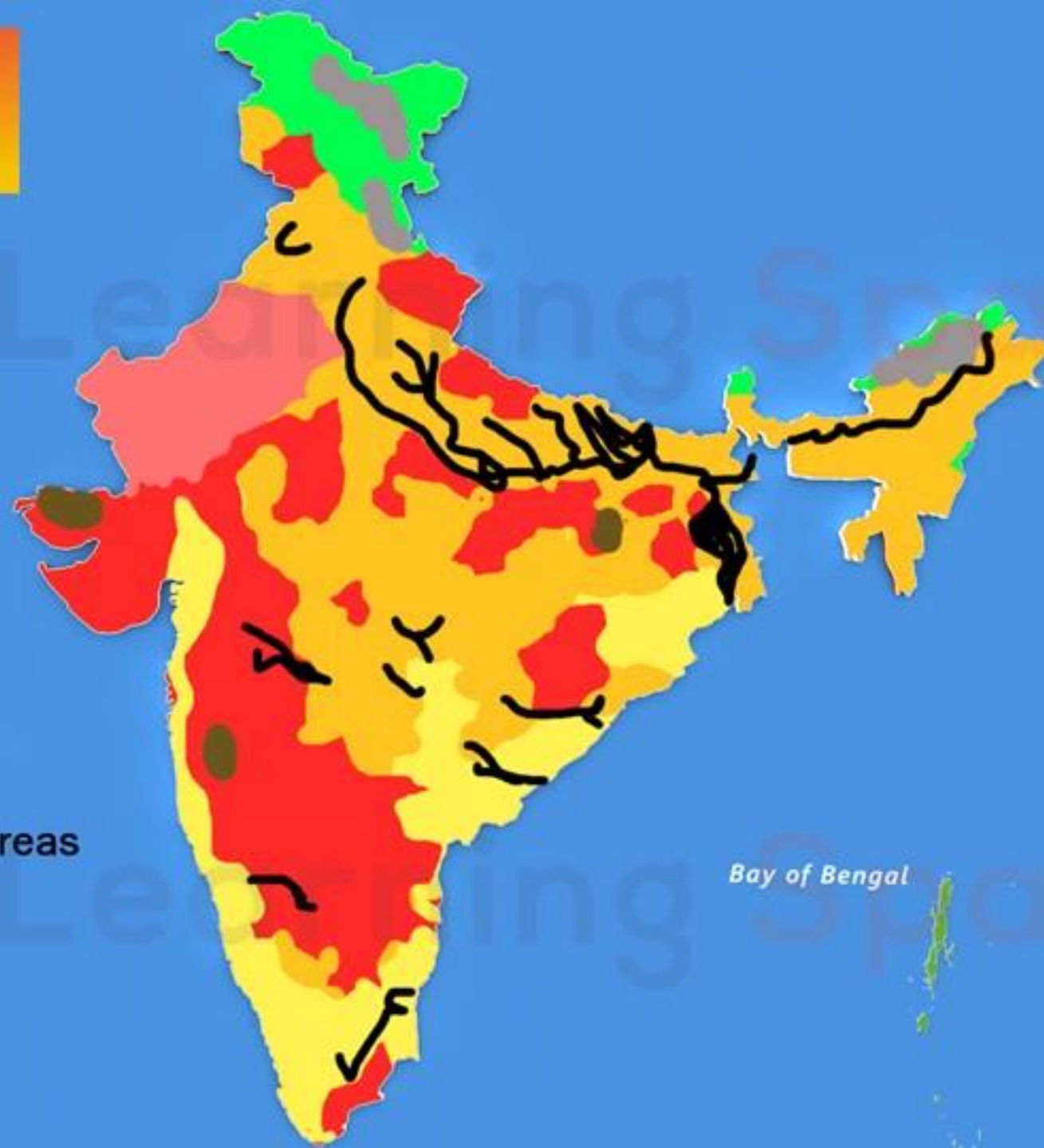
Over 40 million hectares (12%) of its land is prone to floods and river erosion.

Close to 5,700 kms, out of the 7,516 kms long coastline is prone to cyclones and tsunamis.



VULNERABILITY PROFILE OF INDIA

-  Drought prone
-  Cyclone prone areas
-  Hot Desert
-  Cold Desert
-  Floods
-  Earthquakes



CAPACITY BUILDING IN DISASTER MANAGEMENT

... WHY DOES IT MATTER?

OVERVIEW

- 1** Capacity refers to all the **strengths, attributes and resources available** within a community, organization or society to manage and reduce disaster risks and strengthen resilience.
- 2** Like **vulnerability**, capacity depends on **social, economic, political, psychological, environmental and physical assets** and the wider governance regimes.
- 3** The starting point for capacity development is the existing knowledge, strengths, attributes and resources individuals, organizations or society has.

SIGNIFICANCE OF CAPACITY

- 1** Capacity (or the lack of it) is central to reducing disaster risk and therefore critical to meeting development objectives.
- 2** In many low- and middle-income countries, the impacts of regular disasters (extensive risks) are often absorbed by low-income households, thereby maintaining and increasing poverty and undermining development outcomes.
- 3** Enhancing capacity offers vulnerable communities the opportunity to reduce their disaster risk, develop and adapt to climate change



DISASTER MANAGEMENT

BUILDING CAPACITY

COMMUNITY LEVEL

Community capacity building must be used to develop public administrators that are responsive and accountable.



ORGANISATIONAL LEVEL

- Policies and frameworks
- On site job training
- Fund raising
- Training Centres
- Consultation centres

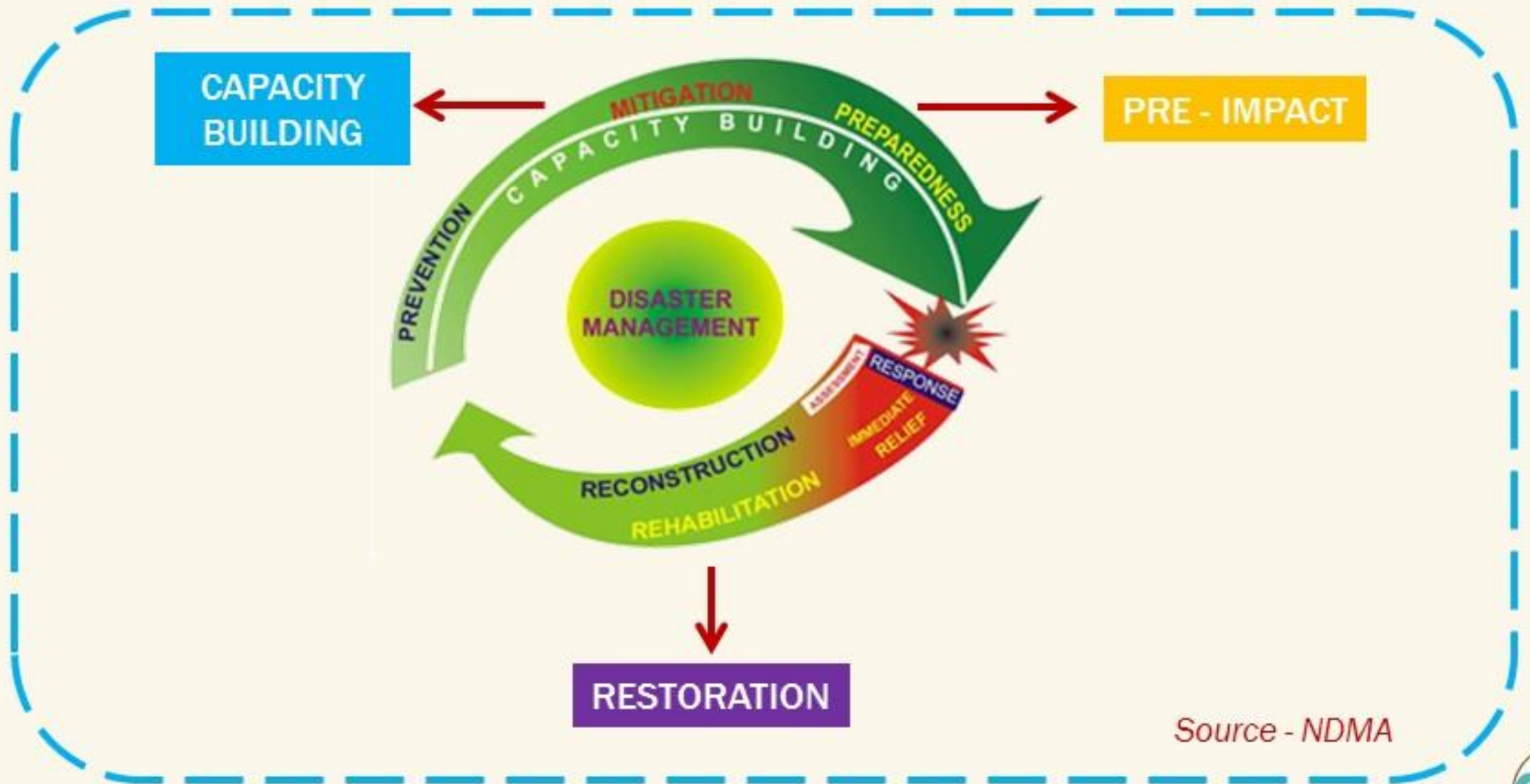


INDIVIDUAL LEVEL

The individual participants should build and enhance knowledge and skills, process of learning and adapting to change.



THE DISASTER MANAGEMENT CYCLE



Source - NDMA



THE VITAL PART OF DISASTER MANAGEMENT

DISASTER RISK REDUCTION

The concept and practice of reducing disaster risks through systematic efforts



DISASTER RISK REDUCTION

PRE-DISASTER RISK REDUCTION

CAPACITY

- Human and infrastructural capital,
- Managing incident response team
- Performing regular mock drills,
- Civil defence,
- Establishing trauma and psychological care centres.



MITIGATION

- Steps to reduce the impacts and risks of hazards
- Adoption and enforcement of land use with respect to hazard mapping
- Implementing or enforcing building codes



PREPAREDNESS

- To take steps to prepare and reduce the effects of disasters.
- Stocktaking of all the exposed capitals.
- Managing capacity and evaluating the risks.
- Local, national and international synergies.



DISASTER RISK REDUCTION

POST - DISASTER RISK REDUCTION

RESCUE

- Providing warning
- Evacuation
- Search and rescue
- Providing immediate assistance.



RELIEF

- Respond to communities who become victims of disaster,
- Relief measures such as food packets, water, medicines, temporary accommodation, relief camps etc.



RECOVERY

- Overall recovery of victims of disaster,
- Recovery of damaged infrastructure and
- Repair of the damages caused etc.



Recovery is the restoration, improvement of facilities, livelihoods and living conditions of disaster affected communities.

EARLY TERM

Duration: 3-18 months

Cash for work, resumption of markets, commerce and trade, restoration of social services, transitional and temporary shelters

MID TERM

Duration: upto 5 years

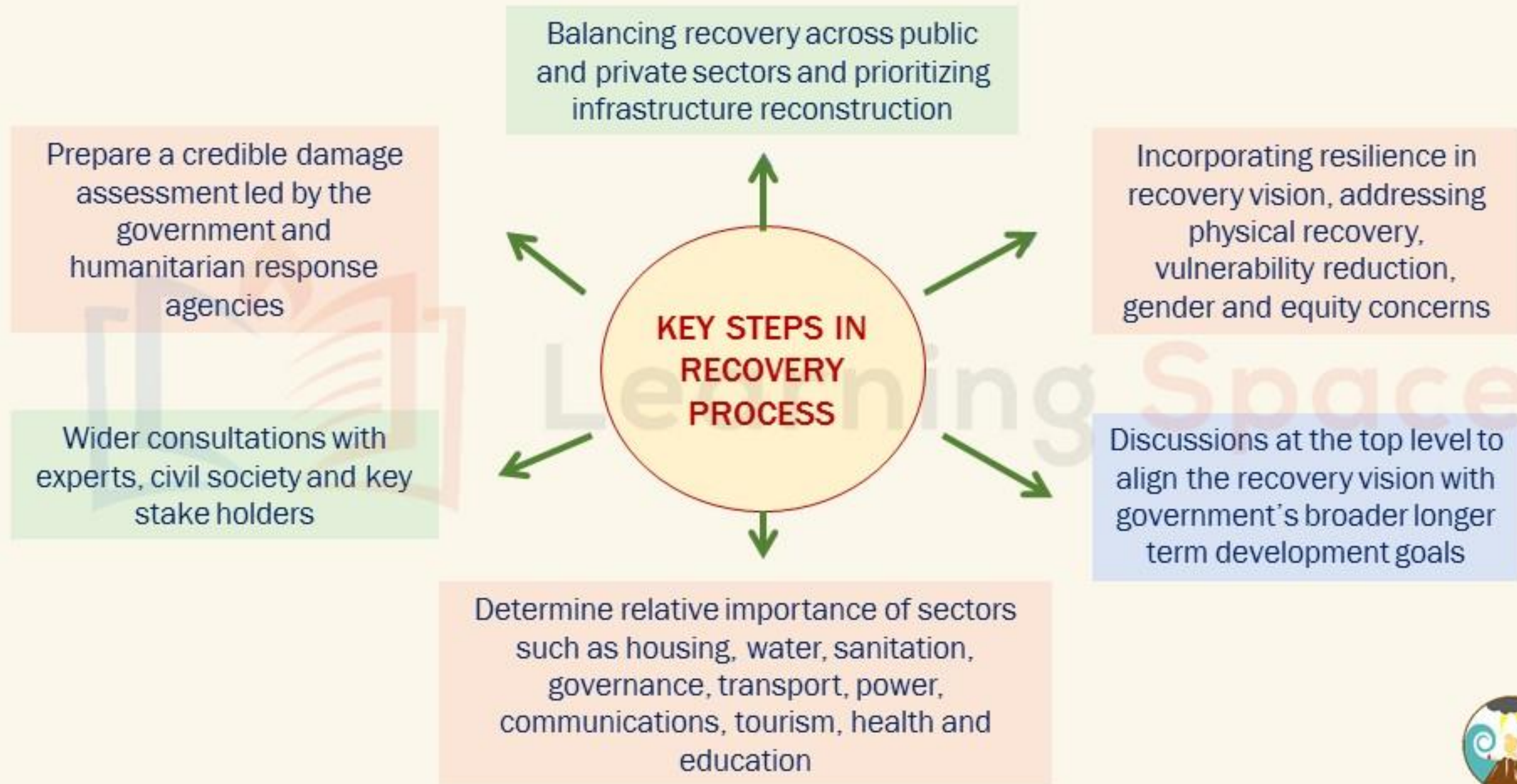
Recovery plans for assets and livelihoods, reconstruction plans for housing, infrastructure, public buildings and cultural heritage buildings

LONG TERM

Duration: within 10 years

Implemented along with development plans: infrastructural strengthening, environmental, urban and regional planning







REHABILITATION MANAGEMENT

A strategy of institutional reform and infrastructural improvement aimed at supporting affected populations

PHYSICAL



SOCIAL



ECONOMIC



DISASTER MANAGEMENT

REHABILITATION MANAGEMENT

PHYSICAL

- All short term and long term rehabilitation of physical infrastructures
- Rehabilitation of agriculture, artisans and animal husbandry
- Adherence to land use planning, flood zone planning, **retrofitting and strengthening of houses**

SOCIAL

- A special looking at the vulnerable groups like elders, young children, artisans, orphans, single women, disabled etc.
- Revival of educational activities and creating awareness
- Promoting self help groups **to create self resilience**

ECONOMIC

- Livelihood restoration, ensuring the continuity of trade and commerce
- Analysis of existing livelihoods strategies and sustainability of businesses
- Accessing to external influencers in imparting skills and knowledge

Another subtype is Psychological rehabilitation to counsel and provide trauma care and therapy



DISASTER RISK REDUCTION

PRIME MINISTER'S TEN-POINT AGENDA FOR RENEWING EFFORTS TOWARDS DISASTER RISK REDUCTION



1

All development sectors must imbibe the principles of disaster risk management

2

Building on local capacity and initiative

3

Working towards risk coverage for all, which starts from poor households to small and medium enterprises, as well as multi-national corporations to nation states.

4

Encourage involvement and leadership of women as they are the biggest sufferers of any disaster.

5

Global investment in risk mapping.



DISASTER RISK REDUCTION

PRIME MINISTER'S TEN-POINT AGENDA FOR RENEWING EFFORTS TOWARDS DISASTER RISK REDUCTION



- 6 Network of universities should be developed to work on disaster issues.
- 7 Leveraging on technology to enhance the efficiency of the disaster risk management efforts
- 8 Opportunity to learn from a disaster must not be wasted
- 9 Greater cohesion in international response to disasters.
- 10 Utilize the opportunities provided by the social media and mobile technologies





RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE



- 1** Discuss the need for disaster risk reduction in India. What are the challenges involved related to government measures. Suggest some ways forward.
- 2** What is capacity and capacity building? How can effective capacity building enable the mitigation of losses both in terms of lives and infrastructure as an effective response strategy?
- 3** Rehabilitation of human settlements is one of the important environmental impacts which always attract controversy while planning major projects. Discuss the measures suggested for mitigation of this impact while proposing major development projects (UPSC 2016)
- 4** Vulnerability is an essential element in defining disaster impacts and threat to people. Discuss different types of vulnerability with reference to disasters. In what ways vulnerability and disasters be characterized? (UPSC 2019)



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